

Short Answer Questions (5 × 2 = 10 Marks)

1. Define a variable in Python. Give one example.

Answer:

A variable is a name used to store a value in Python.

```
age = 20
```

Here, age is the variable and 20 is its value.

2. Write any five Python keywords.

Answer:

```
if
else
for
while
def
```

3. Explain type casting with two examples.

Answer:

Type casting means converting one data type into another data type.

Example 1:

```
a = "10"
b = int(a)
```

Here, string "10" is converted to integer 10.

Example 2:

```
x = 10
y = str(x)
```

Here, integer 10 is converted to string "10".

4. What is the difference between a list and a tuple?

List	Tuple
Uses []	Uses ()
Mutable	Immutable
Can be changed	Cannot be changed
Example: [1, 2, 3]	Example: (1, 2, 3)

5. What is the difference between a set and a dictionary?

Set	Dictionary
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Stores unique values	Stores key-value pairs
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Uses {}	Uses {key: value}
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Example: {1, 2, 3}	Example: {"name": "Arun"}
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Predict the Output (5x2 = 10 Marks)

1. Predict the output

```
x = 10
```

```
y = 5
```

```
print(x + y)
```

```
print(x * y)
```

Answer:

15

50

2. Predict the output

```
print("Hello", name)
```

Answer:

Hello Python

3. Predict the output

```
a = "10"
```

```
b = int(a)
```

```
print(b + 5)
```

Answer:

15

4. Predict the output

```
numbers = [10, 20, 30]
```

```
print(numbers[1])
```

Answer:

20

Why? Python list indexing starts from 0.

```
numbers[0] → 10
```

```
numbers[1] → 20
```

```
numbers[2] → 30
```

5. Predict the output

```
student = {"name": "Arun", "age": 20}
print(student["name"])
```

Answer:

Arun

Practical Coding Assessment

1. Question:

Get student name, age and course using input(). Convert age to int and display the details using print().

Answer:

Input:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
course = input("Enter your course: ")
```

```
print("Name:", name)
print("Age:", age)
print("Course:", course)
```

Output:

Name: Nisha

Age: 23

Course: GEN AI

2. Question:

Get two numbers from the user and display addition, subtraction, multiplication, division, floor division and modulus.

Answer:

Input:

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
```

```
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
```

```
print("Floor Division:", a // b)
print("Modulus:", a % b)
```

Output:

```
Addition: 58.0
Subtraction: -12.0
Multiplication: 805.0
Division: 0.6571428571428571
Floor Division: 0.0
Modulus: 23.0
```